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Cyber Ethical Dilemmas

Cybersecurity is an ever changing and increasingly relevant field. In today's day and age cyber security is more important than ever. Threats loom around every corner of the web. Cyber security is used to enhance privacy and safety online, but is it really? In recent news governments across the globe have been using facial recognition to identify suspects, criminals, and terrorist. This action seems safe, but it raises some questions. Do companies have the people's best interest in mind when developing this product, and are they following ethical and moral values? In theory cybersecurity is meant to protect people's privacy, as both a right and an ethical value, so what does the government think of when using facial recognition technology?

Facial recognition in government practices has been happening under our noses for a while and was recently brought to light in the past few years. This technology is used in investigations, terrorism, and criminal suspect cases. When it comes to scanning the faces of the people to find a criminal the idea does not sound too bad, but when we dive a bit deeper into the data, pictures, and implementations of the software, it gets a bit darker. Facial recognition has actually been in use by law enforcement and government agencies for over a decade, but its scope and prevalence have only recently become widely recognized.

In the United States, for instance, the Federal Bureau of Investigation (FBI) integrated facial recognition into its Next Generation Identification system, which reportedly contains hundreds of millions of images, including driver's license photos. Customs and Border Protection (CBP) has used similar

technology at airports to verify travelers' identities. Internationally, countries like China have adopted large-scale facial recognition for public surveillance, leading to concerns about mass monitoring of citizens. While the immediate value of identifying wanted criminals or terrorism suspects seems obvious, studies—such as a 2018 MIT report—have shown that many facial recognition algorithms struggle with accuracy for individuals with darker skin tones, raising concerns about wrongful arrests or discriminatory policing.

Facial recognition is being implemented into federal law enforcement and our daily lives. While it's ability may seem useful for the time being, catching criminals, it raises some concerns. The fact that this system has been in place for over 10 years and is only gaining concern now is a credit to the secrecy of the government. With this being kept behind curtains it raises the potential idea of Mass Surveillance here in the United States, like it is in China. "It is my hope that this bipartisan report will help inform public policy that will address the myriads of issues concerning artificial intelligence in general, but as it relates to this issue, facial recognition technology specifically. Our country has a moral and legal obligation to ensure that the civil rights and civil liberties of all Americans are protected." [Commissioner Gilchrist]. Not only is mass surveillance a problem, but the recognition is proven to have racial bias of around 100x when identifying African American's, Washington Post.

From a Kantian perspective, people must always be treated as ends in themselves, rather than just mere means, this raises a couple of ethical dilemmas in Kantian ethics. In the context of facial recognition, this violates a duty to respect others' privacy and personal space. Collecting data or scanning faces to compare criminals is against the Kantian code. Furthermore, documented racial bias in facial recognition violates Kants universal moral law, by disadvantaging certain groups over others. As a cybersecurity professional, we need to rigorously advocate for fair algorithms, privacy, and proper ethical code within our systems.

From a utilitarian standpoint, the main idea is to maximize wellbeing with reduced harm. From this standpoint it looks like facial recognition in law enforcement would help a lot. It would reduce

criminal numbers which increases wellbeing in America. However, widespread implementation can provide a couple of concerns from a Utilitarian standpoint. If the algorithm implemented has a racial bias, then it is not maximizing the wellbeing of the population, but only certain portions of it. With that being said if a limited amount of facial recognition were to be used so that the benefit outweighs the harm, then a utilitarian standpoint could prove to be a good choice.

From a Virtue Ethics point of view, moral values are centered around the character and the intentions of the individuals involved. In the case of facial recognition, a virtuous cyber security professional should implement virtues of honesty, justice, and compassion into facial recognition. If a cyber security professional were to do so, then the algorithm would uphold human dignity and societal values. When following virtue ethics, a cyber security professional should consider empathy and compassion to follow the least harmful principle, to inflict the least amount of harm. This commitment to moral character compels a cyber security professional to advocate for transparent, unbiased data collection and responsible oversight, ensuring that any security measures remain consistent with the happiness of both individuals and society at large.

In comparison of all three, Kantian ethics stresses respecting autonomy, so gathering biometric data without consent or perpetuating bias violates individual dignity. Utilitarianism weighs overall well-being, suggesting facial recognition's benefits must exceed the harm it may cause, such as privacy invasions or discrimination. Virtue ethics focuses on moral character, urging cybersecurity professionals to practice justice, honesty, and empathy. Together, these perspectives call for transparent implementation and oversight that respects human rights, balances societal gains, and fosters ethical conduct.

In my opinion the best option for this dilemma is Virtue Ethics. Virtue ethics provides a humanistic empathy to the situation by applying justice, compassion, and honesty to the algorithm. Using virtue ethics would ensure that the algorithm is honest, which would reduce racism and inaccurate results of the facial recognition. This would mean the algorithm is created to do the best it can for right intentions,

catching criminals. Virtue ethics is all around a good choice to approach this situation. One could argue that virtue ethics lacks spine, lacks a clear goal, or even lacks applicable rules. People can claim that it relies on too much individual character, people need to be a good person for it to work. I would say that it does take correct intentions to craft a virtue-oriented algorithm, but necessarily a good person. When it comes to facial recognition, if the person making it has good intentions without harming others, the algorithm would be good.

In conclusion, while both Kantian and utilitarian approaches offer valuable insights, virtue ethics ultimately provides a more honest framework for addressing the ethical challenges of facial recognition. By emphasizing personal moral character, empathy, and the consistent practice of virtues, it encourages cyber security professionals to prioritize human dignity and the greater good in their decisions. This character-driven approach not only helps to reduce issues like privacy violations and racial bias but also builds a commitment to continuous reflection and moral responsibility, a necessity in a rapidly evolving technological world.